

What a Perfect Forecast Still Cannot Tell You

Supply chain is applied economics. Five recurring decisions, priced in dollars on both sides, and what changes once the forecast stops being treated as the answer.

ADAM DEJANS JR. · BIT BROS LLC · AUGUST 2026

EXECUTIVE CONCLUSION

Every buy, every build, every allocation commits capital against a future nobody has seen. The only coherent way to judge such a commitment is in money: what it earns, what it ties up, and what it costs when it turns out wrong in each direction.

Not percentage points of forecast accuracy. Not fill rate. Not turns. A percentage carries no units, and no business banks percentages. The decisions below were all being governed by ratios, and in each one the ratio was hiding a number that turned out to be large, knowable, and in the wrong place.

01	02	03
Price both directions of being wrong before improving the estimate of what will happen.	A service level is a result the economics produce, never a policy chosen in advance.	Under a constraint, rank every candidate unit by marginal value and cut where capacity ends.

On the evidence

Five recurring decisions are drawn on here, from apparel, aftermarket distribution, perishable foods, private-label manufacturing, and aviation sustainment. Company names are invented, identifying details altered, every figure modified, and each case draws on more than one engagement.

The material overlaps with a forthcoming book, *Beyond the Forecast*, which works the same decisions at length. This paper takes the argument that survived contact with all five.

1. The question a forecast does not answer

If tomorrow's forecast were exactly right, which decisions would change?

Fewer than expected, and none of the hard ones.

A regional distributor held a weekly co-packer reservation of 42,000 units on a private-label snack line against 47,000 units of requests from four distribution centers. Suppose the demand signal from every region were perfect. Nothing about that perfection says who gets cut. Describing what will happen and deciding what to do about it are separate questions, and only the first one is a forecasting problem.

The second question is economic, and it stays economic no matter how good the estimate becomes. The four regions differed in what a shortfall cost and what an excess cost: nine dollars against four in the Great Lakes, seven against five in the Mid-Atlantic, eleven against three in the Carolinas, six against six on the Gulf Coast. Those eight numbers decide the allocation. The forecast does not appear in them.

This is the clearest available test of whether an analytics program is aimed at the right target. Take the recurring decision, grant the team a flawless forecast as a thought experiment, and ask what

they would do differently. Where the answer is thin, the constraint sits downstream of the prediction, and more accuracy buys very little.

BOTTOM LINE. A forecast is an input to a decision and never the decision. When the hard part of a choice survives a perfect forecast intact, the hard part was always the economics.

2. Two identical misses, twice the money

What did that forecast error cost?

It depends entirely on which direction it went, and the accuracy metric does not ask.

Take two misses on the same product, each 700 units, each scoring an identical 8.3 percent. In the first, the buy exceeded realized demand by 700 units, tying up roughly \$21,700 in capital that comes back at a loss when it liquidates. In the second, demand outran the buy by the same 700 units, costing roughly \$46,200 in margin never captured, because no inventory remained to sell against the demand that showed up.

Two forecasts, indistinguishable in every statistical sense the organization measures, differing by more than two to one in the money at stake. A planning review reporting MAPE rates both as the same miss. A review asking what each miss cost flags the second as more than twice as expensive, and notices that the two directions are not mirror images for this product.

The distinction worth installing is between statistical performance and decision performance. Statistical performance is how close the forecast came. Decision performance is how much money the resulting decision made or lost. Accuracy metrics earn their place as diagnostics, and they mislead as objectives, because a forecast

can improve on MAPE while the decisions taken from it get worse.

BOTTOM LINE. Report the cost of error in currency alongside the size of error in percent, and the conversation moves to the decision within one review cycle.

3. The service level is an output

Where did the target come from?

In four of the five cases, from nobody currently employed.

An apparel brand committed one winter buy of a single colorway five months ahead of the season. Wholesale \$118, landed cost \$52, liquidation value \$21. That gives \$66 for every unit short against \$31 for every unit long, a ratio putting the economically correct order at the 68th percentile of the demand distribution, near 9,600 units against a point forecast of 8,400.

Nobody had computed it, because the buy was set by ordering to the forecast and adding judgment. The 68 comes out of the four prices. It is a result, not a policy.

An aviation sustainment depot showed the same structure at scale. Tens of thousands of line items, all managed to a single 95 percent availability target. A 75-cent fastener costs 19 cents a year to hold and grounds an aircraft when it is missing, which puts its economically correct availability at essentially 1.0000. An \$18,000 avionics display costs \$4,500 a year to hold and can be expedited for \$2,000, which puts its ratio at 0.31. One target was covering both, and when we asked who owned the 95, the answer was that nobody did.

A target written into a customer contract is a different object and stays legitimate. It constrains the answer rather than replacing the objective, and it should be priced as the constraint it is.

BOTTOM LINE. A uniform target asserts that every item in the catalog carries the same cost of failure. Compute the ratio for a dozen representative items and the assertion collapses without argument.

CASE CARD · APPAREL, ONE SEASONAL BUY	
Decision	One committed quantity, five months ahead of season, no reorder.
Costs	\$66 per unit short (margin never captured); \$31 per unit long (landed cost less liquidation).
Ratio	$66 / (66 + 31) = 0.68$. The buy belongs at the 68th percentile.
Constraint	Mill minimum lot near 1,200 units, then 500-unit factory multiples.
Outcome	Unconstrained optimum near 9,600 lands on 9,500 once the grid is applied. Point forecast was 8,400.

4. A constraint turns a number into a list

How should scarcity be divided?

By ranking every candidate unit and drawing one line.

Return to the 42,000 units against 47,000 requested. The instinct is to cut everyone by the same 10.6 percent, which feels fair and performs badly, because it takes the same proportion from the region whose marginal unit is worth the most as from the region whose underage and overage costs are identical and whose marginal unit is worth the least.

The alternative stops treating this as an allocation. Score every individual candidate unit by its marginal expected value, sort the whole list, and draw the line where capacity runs out. On these numbers that produced 13,100 units to the Great Lakes, 9,500 to the Mid-Atlantic, 9,400 to the Carolinas, and 10,000 to the Gulf Coast, worth roughly \$9,200 in a single week against the even cut.

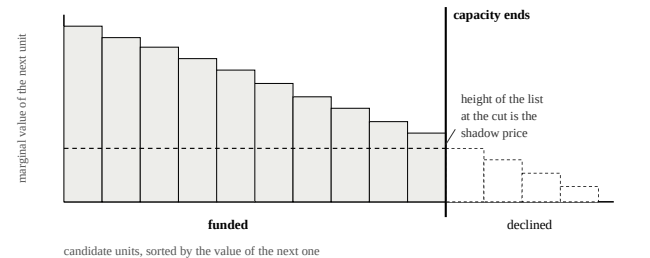


Figure 1. One list, one line. Every unit above the cut was funded for a reason nameable in a sentence. Every unit below it was declined for that same reason stated in reverse. An even-handed percentage cut can say neither.

The height of the list at the cut is the shadow price of the constraint, which converts a planning argument into a procurement question: this is what the next unit of co-packer capacity is worth, so go and see what it costs.

Minimum order quantities do the same work in the other direction. The apparel buy above wanted 9,600 units and could only be placed on a grid of 500s above a mill minimum, so the real choice was between 9,500 and 10,000, and the arithmetic that picked 9,500 is the same arithmetic that ranked the four regions.

BOTTOM LINE. Constraints replace a continuous choice with a short list of feasible ones. Rank the list in dollars and the constraint stops being a complaint and starts being a price.

5. What the component metric and the average conceal

Is 92 percent fill rate good?

Not when three parts have to arrive together.

An aftermarket drivetrain distributor measured availability item by item and reported 92 percent across the three components of a clutch job: pressure plate, friction disc, release bearing. A technician needs all three at once. Under independence those three rates multiply to a 78 percent chance of completing the job, and they are not independent, so the real figure sits lower. The component metric was accurate and the job-level experience was the one the customer had.

Their ocean lead time carried the same shape. Nominal 90 days, actual range 75 to 140, with the long end concentrated rather than scattered. A number stated as a nominal hides that the variability is where the money is.

Averages conceal a second way. A perishable line ran an expected gain of about \$14,400 a year on a policy change, alongside a worst five percent that lost \$107,000 against a single anchor account worth \$310,000 in annual revenue. A business lives through one outcome rather than the average of many, and a decision can carry healthy expected value next to a downside it could not absorb.

BOTTOM LINE. Measure at the level the customer experiences and price the tail separately from the mean. Both failures come from reporting a number that is technically correct at the wrong altitude.

6. What carries the capability afterward

Why do these programs stall after a successful pilot?

They rarely get cancelled. They get orphaned.

Four things carry the work past the pilot team. An owner with suppliers of data and assumptions arranged around them. Recommendations that can be reconstructed later, with overrides

logged rather than discarded, so the reasoning survives the week it was made. Incentives pointed at decision quality. And a written answer to who may change a cost assumption, because that is the question that arrives first and has no natural owner.

The incentive point is the one that gets deferred and should not be. A bonus plan tied to forecast accuracy pays people to protect a metric this argument is trying to demote. Where that plan is still in place, it is the first thing to change rather than the last.

A good decision and a good outcome are different things. The apparel buy above sold ahead of the median and the extra thousand units moved at full price, which is one draw from the distribution and precisely why it is not the argument. The argument is the arithmetic that produced the number five months earlier, and it would have been the same arithmetic, and the same right answer, in a weak season that sent the last five hundred units to liquidation.

BOTTOM LINE. Judge the process that produced the commitment, not the season that followed it, or the organization will learn to make defensible decisions instead of good ones.

THE FIVE DECISIONS

Decision	Governed by	What the ratio was hiding	What replaced it
One seasonal apparel buy	Point forecast plus judgment	\$66 short against \$31 long, putting the buy at the 68th percentile rather than the mean	Critical ratio computed from four prices, then placed on the lot-size grid
Weekly co-packer allocation	Equal-percentage cut across four regions	Marginal value differing nearly fourfold between the best and worst region	One ranked list of candidate units, cut at capacity, shadow price read off the cut
Depot spares availability	A single 95% target, unowned	Ratios running from 0.31 to essentially 1.0000 inside one catalog	Availability derived per item from holding cost, expedite cost, and consequence
Multi-component job fill	Item-level fill rate at 92%	A 78% job completion rate under independence, worse once correlated	Measurement moved to the job, and lead time treated as a range
Perishable line policy	Expected annual value	A worst-five-percent loss of \$107,000 against a \$310,000 account	Tail priced separately and tested against what the business could absorb

FIELDNOTE · QUESTIONS FOR YOUR OWN NUMBERS

Pick one recurring decision worth real money. What does one unit short cost, in currency? What does one unit long cost? Who last computed those two numbers, and when? If the forecast for this decision were perfect tomorrow, what would you do differently? If the honest answer is very little, the next project is not a forecasting project.

7. Where to start

Converting every decision at once is the most reliable way to convert none. The screen that has worked runs on six criteria in three pairs. Economic: the stakes involved, and how much genuine uncertainty surrounds them. Modeling: how often the decision repeats, and how tractable it is. Organizational: how much implementation burden it carries, and whether it has an owner.

Five of those are graded low, medium, or high. Ownership is a gate. A decision with high stakes, ample data, and no accountable

owner will not convert, and grading ownership on the same scale as the other five hides that.

Ninety days is enough to convert one decision and to establish that the method works. It requires a baseline measured before anything changes and a failure standard written down before any evidence arrives, ideally by the person most skeptical that the method will work. A pilot without a written failure standard will be argued about rather than concluded.

8. A maturity model built to be thrown away

How would we know we were making progress?

By what the organization can put a dollar figure on, and nothing else.

Most maturity models grade the possession of techniques, which rewards buying software. The one below grades a single capability at five depths, refuses to name any level after a method, and can be checked against the accounts.

WHAT THE ORGANIZATION CAN PRICE

Level 1	Nothing, in currency. Performance is discussed in percentages, and the cost of a miss is described as significant rather than counted.
Level 2	One side of the error, once. Usually the visible side: excess that had to be written down, or expedite freight that showed up in an accrual.
Level 3	Both sides of one unit of error, for a decision that matters. This is the threshold at which a service level becomes an output.
Level 4	Every feasible decision, and the constraint itself. The ranked list exists, and the height at the cut is quoted to procurement as a price.
Level 5	The decisions it did not make, every cycle. Declined options are priced and logged, so the cost of the road not taken is visible rather than assumed away.

The test is worth stating in the harshest available form. If an organization can move from one level to the next without a single figure on its profit and loss statement changing, the model is decoration and should be thrown away.

Movement between levels is not primarily technical. Getting from the second to the third means someone with authority signs their

name to a cost of shortage, and that signature is usually the binding constraint rather than the arithmetic. Five moves make it easier: admit the loss the current rule is producing, locate whoever actually holds the authority to change it, report the result in the units that person already uses, let the most skeptical person in the room write the failure standard, and publish the losses alongside the wins.

THE CORE TAKEAWAY

Supply chain is applied economics. Price both directions of being wrong on one decision that matters, and most of the instruments currently standing in for that arithmetic will retire themselves.

ABOUT BIT BROS

Bit Bros is a boutique applied economics firm working on supply chain and decision systems, at the intersection of optimization, simulation, forecasting, and AI. Founder experience includes Amazon, Toyota, Gurobi, and AT&T. We work on the operating mechanism around the model: how decisions get priced, ranked, tested, owned, and revisited.

RELATED READING

Beyond the Forecast: A Leader’s Introduction to Supply Chain Uncertainty works these five decisions at length, along with the maturity model and the ninety-day sequence sketched above. *The Decision Factory* (DeJans and Elam, 2026) puts the same argument in narrative form.

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